

Exponential Stability, Passivity, and Dissipativity Analysis of Generalized Neural Networks With Mixed Time-Varying Delays

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Abstract—In this paper, we analyze the exponential stability, passivity, and $(\Omega, \mathcal{S}, \mathcal{R})$ - γ -dissipativity of generalized neural networks (GNNs) including mixed time-varying delays in state vectors. Novel exponential stability, passivity, and $(\Omega, \mathcal{S}, \mathcal{R})$ - γ -dissipativity criteria are developed in the form of linear matrix inequalities for continuous-time GNNs by constructing an appropriate Lyapunov–Krasovskii functional (LKF) and applying a new weighted integral inequality for handling integral terms in the time derivative of the established LKF for both single and double integrals. Some special cases are also discussed. The superiority of employing the method presented in this paper over some existing methods is verified by numerical examples.

Index Terms—Exponential passivity, generalized neural networks (GNNs), $(\Omega, \mathcal{S}, \mathcal{R})$ - γ -dissipativity, time-varying delay, weighted integral inequality (WII).

I. INTRODUCTION

ARTIFICIAL neural networks (NNs) have become a very popular research theme in the past two decades, and they are broadly studied, since they have many applications in several areas, such as image processing, communication, fault diagnosis, fixed-point computations, parallel computations, and industrial automation [1]–[9]. The applications of NNs usually depend on the behavior and stability characteristic of the system. Thus, it is worth analyzing the stability of NNs. When implementing artificial NNs, time-delayed signals often

occur due to the inherent communication time between the finite switching speed of amplifiers and neurons, which can lead to divergence, oscillation, and even instability. Thus, the stability problems of time-delayed NNs have become a major subject of research, and many interesting results have been proposed in recent years [10]–[25].

Xu *et al.* [26] stated that artificial NNs can be classified into two NN models: 1) scilicet and static NNs [27] and 2) local-field NNs [28]. Local-field NNs can be designed in various forms, such as cellular, Cohen–Grossberg, bidirectional associative memory (BAM), and Hopfield NNs. For instance, the stability of Hopfield NNs was investigated in [12]. Cheng *et al.* [13] studied the existence and stability problem of delayed discrete-time Cohen–Grossberg NNs. To obtain the uniqueness and global exponential stability, M-matrix theory and the contracting mapping principle were used. In [28], the passivity problem was studied for stochastic neutral-type memristor-based BAM NNs including mixed time-varying delayed signals, where the authors introduced the quadruple integral Lyapunov–Krasovskii functional (LKF) for stochastic stability. Compared with local field NNs, only a small number of stability problems have been studied for static NNs [27], [29]. For example, the improved stability of static NNs including interval time-delayed signals is presented in [27]. In [29], the global exponential stability problem was studied for delayed static NNs. Moreover, local-field and static NNs can be changed from one form to an equivalent form by introducing a few assumptions. However, these assumptions cannot be applied to numerous practical applications [9]. Therefore, the above NNs are not the same, and the two NN models are not always transferred equivalently. As a result, the stability problems for these two NN models are generally studied separately. Very recently, a unified model called a generalized neural network (GNN) was constructed by combining these two NN models (see [30]–[33]). Currently, many researchers are focusing their research on GNNs due to their variety of applications in biological networks, associative memories, signal processing, the tasks of pattern recognition, and other scientific areas.

Analyzing the stability of NNs has become the center of attention for various research proceedings. In recent years, asymptotic, complete, and global exponential stability problems have been investigated for NNs [29], [33]. As discussed in [34], the investigation of the exponential

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stability criteria is very significant, while the exponential convergence rate can be utilized to ensure the speed of neural calculations. Thus, studying the problem of exponential stability has some practical importance, and it is useful to estimate and ensure the exponential convergence rate of NNs. Therefore, analyzing exponential stability problems for delayed NNs has become important in both theory and practice [35]–[41].

Recently, the dissipativity problem has gained extensive research attention, since it is an important subject for physical systems, such as the design of group coordination, hybrid systems, nonlinear time-delay systems, and Internet-based control [42]–[48]. Passivity, and its generalization dissipativity, characterizes the energy consumption of a system and is used in a variety of applications (e.g., electrical, mechanical, chemical, and communication systems). In most cases, passivity can be used to demonstrate that passive systems will be stable under specific criteria [49]. Passivity is a section of a general and broader theory of dissipativeness. That is, the dissipativity criteria are a more general case of passivity and stability analyses. Thus, it is essential to consider dissipativity problems for NNs. In [43], the global dissipativity problem for delayed recurrent NNs was studied. In [44], the dissipativity problem was investigated for uncertain inertial NNs. In [45], the dissipativity criteria of delayed memristive NNs with randomly occurring uncertainties were established. In [46], the synchronization of uncertain chaotic NNs with actuator failures was studied based on extended dissipativity criteria. Strict dissipativity and stability problems were analyzed for direct-form digital filters in [50]. To the best of the authors' knowledge, the exponential stability, passivity, and dissipativity problems for delayed GNNs have not yet been completely considered in the literature, which partly motivates us to conduct study.

Many methods have been established using various mathematical methods (e.g., Jensen's inequality [51], the free-weighting matrix approach [28], and the delay-partitioning method [52]). Stability investigations have been conducted using some new inequalities (e.g., reciprocal convex combination (RCC) [53], Wirtinger-based inequality [54], and auxiliary function-based inequality [55]). It has been shown that these approaches provide less conservative results for the stability analysis of the problems under consideration and require few decision variables compared with other existing approaches in the literature. However, Hien and Trinh [56] proposed a new weighted integral inequality (WII) for exponential Lyapunov functionals.

Based on the above views, in this paper, the exponential stability, passivity, and $(\Omega, \mathfrak{S}, \mathfrak{R})$ - γ -dissipativity problems for delayed GNNs are considered. A novel improved method is established for the exponential stability of delayed GNNs by using a new WII-based mathematical analysis for both single and double integrals. To demonstrate the effectiveness of the established results, some numerical examples are given. The obtained results are compared with those of recent methods in the literature.

Some notations used in this paper are presented in the following table.

Notation	Description
$\mathcal{R}^n$	The Euclidean space with n -dimensions
$\mathcal{R}^{m \times n}$	The set of $m \times n$ dimension real matrices
$Y > 0$	Y is a positive-definite and symmetric matrix
$\text{diag}\{a_1, \dots, a_n\}$	A block diagonal matrix
$\mathcal{C}([-\tau, 0]; \mathcal{R}^n)$	Set of continuous functions ψ
$\mathcal{L}_{\mathcal{F}}^b([-\tau, 0]; \mathcal{R}^n)$	Mapping from $[-\tau, 0]$ to $\mathcal{R}^n$
$(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathcal{P})$	Set of $\mathcal{F}$ -measurable
$\mathcal{E}(\cdot)$	$\mathcal{C}([-\tau, 0]; \mathcal{R}^n)$ -valued variables
$L_2[0, \infty)$	The complete probability space
	The mathematical expectation
	The space of square-integrable vector functions over $[0, \infty)$

II. SYSTEM FORMULATION

Let us consider GNNs with mixed time-varying delays

$$\begin{aligned} \dot{\varpi}(t) = & -C\varpi(t) + W_0 f(W_3 \varpi(t)) + W_1 f(W_3 \varpi(t - d(t))) \\ & + W_2 \int_{t-\tau(t)}^t f(W_3 \varpi(\varepsilon)) d\varepsilon + z(t) \\ \mathfrak{z}(t) = & f(W_3 \varpi(t)) \end{aligned} \quad (1)$$

where $\varpi(t) = [\varpi_1(t), \varpi_2(t), \dots, \varpi_n(t)]^T \in \mathcal{R}^n$ is the neuron state vector signal. $C = \text{diag}\{c_1, \dots, c_n\} > 0$ is a known diagonal matrix. W_0 , W_1 , W_2 , and W_3 are the interconnection weight matrices of the neurons. $z(t) \in \mathcal{R}^n$ is the inputs belonging to $L_2[0, \infty)$. $\mathfrak{z}(t)$ is the output vector of the NNs. $f(W_3 \varpi(t)) = [f_1(W_3 \varpi_1(t)), \dots, f_n(W_3 \varpi_n(t))]^T \in \mathcal{R}^n$ denotes the activation function of neurons. $d(t)$ and $\tau(t)$ denote time-varying discrete delay and time-varying distributed delay, respectively. They are assumed to satisfy

$$0 < d(t) \leq d, \quad \dot{d}(t) \leq \mu, \quad 0 < \tau(t) \leq \tau \quad (2)$$

where d , τ , and μ are positive constants.

It should be noted that for $i = 1, 2, \dots, n$, $f_i(\cdot)$ satisfies the following assumption.

Assumption 1 [57]: For any $i = 1, 2, \dots, n$, $f_i(0) = 0$ and there exist constant matrices $H_1 = \text{diag}\{m_1^-, m_2^-, \dots, m_n^-\}$ and $H_2 = \text{diag}\{m_1^+, m_2^+, \dots, m_n^+\}$ such that

$$m_i^- \leq \frac{f_i(W_{3i}u) - f_i(W_{3i}v)}{W_{3i}u - W_{3i}v} \leq m_i^+ \quad (3)$$

holds for any $u, v \in \mathcal{R}$ with $u \neq v$.

Definition 1 [59], [60]: The GNN (1) is said to be exponentially dissipative from input $z(t)$ to output $\mathfrak{z}(t)$, and the quadratic supply rate $\gamma(z, \mathfrak{z}) = \mathfrak{z}^T \Omega \mathfrak{z} + 2\mathfrak{z}^T \mathfrak{S} z + z^T \mathfrak{R} z$, where $\Omega \leq 0$, $\mathfrak{R} = \mathfrak{R}^T > 0$, and $\mathfrak{S}$ are real known matrices, if there exists an exponential Lyapunov function (storage function) V , and a constant $\alpha > 0$ such that for any input $z(t)$ and any initial condition $\varpi(t_0)$, the following relation holds:

$$\dot{V}(\varpi(t)) + \alpha V(\varpi(t)) \leq \gamma(z, \mathfrak{z}), \quad t > t_0.$$

Remark 1: The global dissipativity and global exponential dissipativity criteria are established in [43] for Hopfield and cellular NNs with or without time delays. To achieve this, a radically unbounded and positive-definite Lyapunov function has been employed. Then, by solving and using the definition based on the attractive set, the dissipativeness was investigated. However, Definition 1 in this paper states the properties of the $(\Omega, \mathfrak{S}, \mathfrak{R})$ - γ -dissipativity criteria of the GNN (1). By using

the LKF and linear matrix inequality (LMI) techniques, the dissipativeness of the GNN has been analyzed.

Lemma 1 [56]: For a given matrix $R \in \mathcal{R}_{>0}^{n \times n}$, scalar $\alpha > 0$, and function $\chi \in \mathcal{C}([a, b], \mathcal{R}^n)$, satisfy

$$\int_a^b e^{\alpha(s-b)} \chi^T(\varepsilon) R \chi(\varepsilon) d\varepsilon \geq \frac{\alpha}{\bar{\eta}_0} \left(\int_a^b \chi(\varepsilon) d\varepsilon \right)^T R \left(\int_a^b \chi(\varepsilon) d\varepsilon \right) + \frac{\alpha}{\bar{\rho}_0} \zeta^T (K_1^T K_1 \otimes R) \zeta$$

where

$$\begin{aligned} \rho_0 &= \frac{\bar{\eta}_0}{\bar{\eta}_1^2} [\bar{\eta}_0^2 - (\alpha l)^2 e^{\alpha l}], \bar{\eta}_0 = e^{\alpha l} - 1, \bar{\eta}_1 = \bar{\eta}_0 - \alpha l \\ l &= b - a, \zeta = \left[\int_a^b \chi(\varepsilon) d\varepsilon \int_a^b \int_\varepsilon^b \chi(u) du d\varepsilon \right]^T \\ K_1 &= [1 \quad -\alpha \bar{\eta}_0 / \bar{\eta}_1]. \end{aligned}$$

Lemma 2 [56]: For a given matrix $Z \in \mathcal{R}_{>0}^{n \times n}$, scalar $\alpha > 0$, and function $\chi \in \mathcal{C}([a, b], \mathcal{R}^n)$, satisfy

$$\int_a^b \int_\varepsilon^b e^{\alpha(\theta-b)} \chi^T(\theta) Z \chi(\theta) d\theta d\varepsilon \geq \frac{\alpha^2}{\bar{\eta}_1} \psi^T(t) (K_0^T K_0 \otimes Z) \psi(t) + \frac{4\alpha^2}{\bar{\rho}_1} \psi^T(t) (K_2^T K_2 \otimes Z) \psi(t)$$

where

$$\begin{aligned} \rho_1 &= \frac{\bar{\eta}_1}{\bar{\eta}_2^2} [2\eta_1^2 - (\alpha d + (\alpha d - 1)\bar{\eta}_0)(\alpha d)^2] \\ \bar{\eta}_2 &= \bar{\eta}_1 - \frac{(\alpha d)^2}{2}, l = b - a, K_0 = [1 \quad 0] \\ K_2 &= [1 \quad -\alpha \bar{\eta}_1 / \bar{\eta}_2] \\ \psi &= \left[\int_a^b \int_\varepsilon^b \chi(\theta) d\theta d\varepsilon \int_a^b \int_\varepsilon^b \int_\theta^b \chi(\theta) d\theta du d\varepsilon \right]^T. \end{aligned}$$

III. MAIN RESULTS

Define

$$\begin{aligned} \vartheta(t) &= \left[\varpi^T(t) \int_{t-d}^t \varpi^T(\varepsilon) d\varepsilon \int_{-d}^0 \int_{t+\varepsilon}^t \varpi^T(u) du d\varepsilon \right]^T \\ e_\ell &= [0_{n \times (\ell-1)n} \quad I_n \quad 0_{n \times (8-\ell)n}]^T, \ell = 1, 2, \dots, 8 \\ \xi(t) &= \left[\varpi^T(t) \quad \varpi^T(t-d) \quad f^T(W_3 \varpi(t)) \right. \\ &\quad \times f^T(W_3 \varpi(t-d)) \quad \int_{t-d}^t \varpi^T(\varepsilon) d\varepsilon \\ &\quad \times \int_{t-\tau(t)}^t f^T(W_3 \varpi(\varepsilon)) d\varepsilon \quad \left. \int_{-d}^0 \int_{t+\varepsilon}^t \varpi^T(u) du d\varepsilon \quad z(t) \right]^T. \end{aligned}$$

The following theorem has been proposed based on the LKF and LMI techniques.

Theorem 1: Given $\alpha > 0$ and some scalars $d > 0$, $\tau > 0$, $\mu > 0$, the GNN (1) is dissipative when there exist matrices, $P \in \mathcal{R}_{>0}^{3n \times 3n}$ and $Q, R, S, T, Z \in \mathcal{R}_{>0}^{n \times n}$, and diagonal matrices $\Gamma_p \in \mathcal{R}_{>0}^{n \times n}$, ($p = 1, 2, 3$), $\Lambda_q \in \mathcal{R}_{>0}^{n \times n}$ ($q = 1, 2$), such that

$$\sum_{m=1}^8 \Xi_m < 0 \quad (4)$$

where

$$\begin{aligned} \Phi_0 &= \begin{bmatrix} e_1 \\ e_5 \\ e_7 \end{bmatrix}, \Phi_1 = \begin{bmatrix} \mathbb{A} \\ e_1 - e_2 \\ de_1 - e_5 \end{bmatrix}, \Phi_2 = \begin{bmatrix} e_1 - e_2 \\ de_1 - e_5 \end{bmatrix} \\ \Phi_3 &= \begin{bmatrix} de_1 - e_5 \\ \frac{d^2}{2} e_1 - e_7 \end{bmatrix}, \Xi_1 = \Phi_0^T P \Phi_1 + \Phi_1^T P \Phi_0 + \alpha \Phi_0^T P \Phi_0 \\ \Xi_2 &= 2[e_3 - H_1 W_3 e_1]^T \Lambda_1 W_3 \mathbb{A} + 2[H_2 W_3 e_1 - e_3]^T \Lambda_2 W_3 \mathbb{A} \\ \Xi_3 &= e_1^T Q e_1 - e^{-\alpha d} e_2^T Q e_2 + e_3^T [S + \tau^2 T] e_3 \\ &\quad + \pi(\mu) e_4^T S e_4 + \mathbb{A}^T \left[dR + \frac{d^2}{2} Z \right] \mathbb{A} \\ \Xi_4 &= -\left(\frac{\alpha}{\eta_0} (e_1 - e_2)^T R (e_1 - e_2) + \frac{\alpha}{\rho_0} \Phi_2^T (K_1^T K_1 \otimes R) \Phi_2 \right) \\ \Xi_5 &= -(1 - \mu) e^{-\alpha d} e_6^T T e_6 \\ \Xi_6 &= -\left(\frac{\alpha^2}{\eta_1} (de_1 - e_5)^T Z \right. \\ &\quad \times (de_1 - e_5) + \frac{4\alpha^2}{\rho_1} \Phi_3^T (K_2^T K_2 \otimes Z) \Phi_3 \left. \right) \\ \Xi_7 &= -2e_1^T W_3^T H_1 \Gamma_1 H_2 W_3 e_1 + e_1^T W_3^T (H_1 + H_2) \Gamma_1 e_3 \\ &\quad + e_3^T \Gamma_1 (H_1 + H_2) W_3 e_1 - 2e_3^T \Gamma_1 e_3 \\ &\quad - 2e_2^T W_3^T H_1 \Gamma_2 H_2 W_3 e_2 + e_2^T W_3^T (H_1 + H_2) \Gamma_2 e_4 \\ &\quad + e_4^T \Gamma_2 (H_1 + H_2) W_3 e_2 - 2e_4^T \Gamma_2 e_4 - 2[e_3 - e_4 \\ &\quad - H_1 W_3 (e_1 - e_2)]^T \Gamma_3 [e_3 - e_4 - H_2 W_3 (e_1 - e_2)] \\ \Xi_8 &= -e_3^T Q e_3 - 2e_3^T S e_8 - e_8^T S e_8, \mathbb{A} = -C e_1 \\ &\quad + W_0 e_3 + W_1 e_4 + W_2 e_6 + e_8, \eta_2 = \eta_1 - \frac{(\alpha d)^2}{2} \\ K_1 &= [1 \quad -\alpha \eta_0 / \eta_1], K_2 = [1 \quad -\alpha \eta_1 / \eta_2] \\ \eta_0 &= e^{\alpha d} - 1, \eta_1 = \eta_0 - \alpha d, \rho_0 = \frac{\eta_0}{\eta_1^2} [\eta_0^2 - (\alpha d)^2 e^{\alpha d}] \\ \rho_1 &= \frac{\eta_1}{\eta_2^2} [2\eta_1^2 - (\alpha d + (\alpha d - 1)\eta_0)(\alpha d)^2] \end{aligned}$$

and

$$\pi(\mu) = \begin{cases} -(1 - \mu) e^{-\alpha d}, & \mu \leq 1 \\ -(1 - \mu), & \mu > 1. \end{cases}$$

Proof: Let the following LKF be:

$$\begin{aligned} V(\varpi_t) &= \vartheta^T(t) P \vartheta(t) + 2 \sum_{i=1}^n \lambda_{1i} \int_0^{W_{3i} \varpi(t)} (f_i(\varepsilon) - m_i^- \varepsilon) d\varepsilon \\ &\quad + 2 \sum_{i=1}^n \lambda_{2i} \int_0^{W_{3i} \varpi(t)} (m_i^+ \varepsilon - f_i(\varepsilon)) d\varepsilon \\ &\quad + \int_{t-d}^t e^{\alpha(\varepsilon-t)} \varpi^T(\varepsilon) Q \varpi(\varepsilon) d\varepsilon \\ &\quad + \int_{-d}^0 \int_{t+\varepsilon}^t e^{\alpha(\theta-t)} \dot{\varpi}^T(\theta) R \dot{\varpi}(\theta) d\theta d\varepsilon \\ &\quad + \int_{t-d(t)}^t e^{\alpha(\varepsilon-t)} f^T(W_3 \varpi(\varepsilon)) S f(W_3 \varpi(\varepsilon)) d\varepsilon \\ &\quad + \tau \int_{t-\tau}^0 \int_{t+\varepsilon}^t e^{\alpha(\theta-t)} f^T(W_3 \varpi(\theta)) T f(W_3 \varpi(\theta)) d\theta d\varepsilon \\ &\quad + \int_{-d}^0 \int_\varepsilon^0 \int_{t+\theta}^t e^{\alpha(u-t)} \dot{\varpi}^T(u) Z \dot{\varpi}(u) du d\theta d\varepsilon. \quad (5) \end{aligned}$$

It follows from (5) that $V(\varpi_t) \geq \lambda_{\min}(P_{11})\|\varpi(t)\|^2$. The derivative of $V(\varpi_t)$ along the trajectory of (1) can be obtained as follows:

$$\begin{aligned} \dot{V}(\varpi_t) + \alpha \bar{V}(\varpi_t) &\leq \xi^T(t) \left[\sum_{n=1}^3 \Xi_n \right] \xi(t) - \int_{t-d}^t e^{\alpha(\varepsilon-t)} \dot{\varpi}^T(\varepsilon) R \dot{\varpi}(\varepsilon) d\varepsilon \\ &\quad - \tau \int_{t-\tau}^t e^{\alpha(\varepsilon-t)} f^T(W_3 \varpi(\varepsilon)) T f(W_3 \varpi(\varepsilon)) d\varepsilon \\ &\quad - \int_{-d}^0 \int_{t+\varepsilon}^t e^{\alpha(\theta-t)} \dot{\varpi}^T(\theta) Z \dot{\varpi}(\theta) d\theta d\varepsilon \end{aligned} \quad (6)$$

where $\bar{V}(\varpi_t) = V(\varpi_t) - 2 \sum_{i=1}^n \lambda_{1i} \int_0^{W_3 \varpi(t)} (f_i(\varepsilon) - m_i^- \varepsilon) d\varepsilon - 2 \sum_{i=1}^n \lambda_{2i} \int_0^{W_3 \varpi(t)} (m_i^+ \varepsilon - f_i(\varepsilon)) d\varepsilon$. Through Lemmas 1 and 2 and Jensen's inequality, we can obtain

$$- \int_{t-d}^t e^{\alpha(\varepsilon-t)} \dot{\varpi}^T(\varepsilon) R \dot{\varpi}(\varepsilon) d\varepsilon \leq \xi^T(t) \Xi_4 \xi(t) \quad (7)$$

$$- \tau \int_{t-\tau}^t e^{\alpha(\varepsilon-t)} f^T(W_3 \varpi(\varepsilon)) T f(W_3 \varpi(\varepsilon)) d\varepsilon \leq \xi^T(t) \Xi_5 \xi(t) \quad (8)$$

$$\int_{-d}^0 \int_{t+\varepsilon}^t e^{\alpha(\theta-t)} \dot{\varpi}^T(\theta) Z \dot{\varpi}(\theta) d\theta d\varepsilon \leq \xi^T(t) \Xi_6 \xi(t). \quad (9)$$

From Assumption 1, we have

$$0 \leq 2[f(W_3 \varpi(t)) - H_1 W_3 \varpi(t)]^T \Gamma_1 [H_2 W_3 \varpi(t) - f(W_3 \varpi(t))] \quad (10)$$

$$0 \leq 2[f(W_3 \varpi(t-d(t))) - H_1 W_3 \varpi(t-d(t))]^T \Gamma_2 \times [H_2 W_3 \varpi(t-d(t)) - f(W_3 \varpi(t-d(t)))] \quad (11)$$

$$0 \leq 2[f(W_3 \varpi(t)) - f(W_3 \varpi(t-d(t)))]^T \Gamma_3 \times [f(W_3 \varpi(t)) - f(W_3 \varpi(t-d(t)))] \\ - H_1 (W_3 \varpi(t) - W_3 \varpi(t-d(t)))^T \Gamma_3 \times [f(W_3 \varpi(t)) - f(W_3 \varpi(t-d(t)))] \\ - H_2 (W_3 \varpi(t) - W_3 \varpi(t-d(t))). \quad (12)$$

By combining (6) with (12), we obtain

$$\dot{V}(\varpi_t) + \alpha \bar{V}(\varpi_t) - \gamma(z(t), \mathfrak{z}(t)) \leq \xi^T(t) \left[\sum_{m=1}^8 \Xi_m \right] \xi(t) \quad (13)$$

where Ξ_m , ($m = 1, 2, \dots, 8$) are defined in (4). If the LMI (4) holds, then we obtain $\dot{V}(\varpi_t) + \alpha \bar{V}(\varpi_t) \leq \gamma(z(t), \mathfrak{z}(t))$. Then, through Definition 1, the GNN (1) is dissipative. ■

Remark 2: In Theorem 1, the passivity and $(\mathfrak{Q}, \mathfrak{S}, \mathfrak{R})$ - γ -dissipativity problems for the GNN have been considered. To obtain the stability conditions, when $z(t) = \mathfrak{z}(t) = 0$, the GNN (1) becomes

$$\begin{aligned} \dot{\varpi}(t) &= -C\varpi(t) + W_0 f(W_3 \varpi(t)) + W_1 f(W_3 \varpi(t-d(t))) \\ &\quad + W_2 \int_{t-\tau(t)}^t f(W_3 \varpi(\varepsilon)) d\varepsilon. \end{aligned} \quad (14)$$

The following theorem states the stability conditions for the GNN (14).

Theorem 2: Given $\alpha > 0$ and some scalars $d > 0$, $\tau > 0$, $\mu > 0$, the GNN (14) is exponentially stable when there exist matrices, $P \in \mathcal{R}_{>0}^{3n \times 3n}$ and $Q, R, S, T, Z \in \mathcal{R}_{>0}^{n \times n}$, and

diagonal matrices, $\Gamma_p \in \mathcal{R}_{>0}^{n \times n}$, ($p = 1, 2, 3$), $\Lambda_q \in \mathcal{R}_{>0}^{n \times n}$ ($q = 1, 2$), such that

$$\sum_{m=1}^7 \Xi_m < 0 \quad (15)$$

where

$$\begin{aligned} \Phi_0 &= \begin{bmatrix} e_1 \\ e_5 \\ e_7 \end{bmatrix}, \quad \Phi_1 = \begin{bmatrix} \mathbb{A} \\ e_1 - e_2 \\ de_1 - e_5 \end{bmatrix}, \quad \Phi_2 = \begin{bmatrix} e_1 - e_2 \\ de_1 - e_5 \end{bmatrix} \\ \Phi_3 &= \begin{bmatrix} de_1 - e_5 \\ \frac{d^2}{2} e_1 - e_7 \end{bmatrix}, \quad \Xi_1 = \Phi_0^T P \Phi_1 + \Phi_1^T P \Phi_0 + \alpha \Phi_0^T P \Phi_0 \\ \Xi_2 &= 2[e_3 - H_1 W_3 e_1]^T \Lambda_1 W_3 \mathbb{A} + 2[H_2 W_3 e_1 - e_3]^T \Lambda_2 W_3 \mathbb{A} \\ \Xi_3 &= e_1^T Q e_1 - e^{-\alpha d} e_2^T Q e_2 + e_3^T [S + \tau^2 T] e_3 \\ &\quad + \pi(\mu) e_4^T S e_4 + \mathbb{A}^T \left[dR + \frac{d^2}{2} Z \right] \mathbb{A} \\ \Xi_4 &= - \left(\frac{\alpha}{\eta_0} (e_1 - e_2)^T R (e_1 - e_2) + \frac{\alpha}{\rho_0} \Phi_2^T (K_1^T K_1 \otimes R) \Phi_2 \right) \\ \Xi_5 &= -(1 - \mu) e^{-\alpha d} e_6^T T e_6, \quad \Xi_6 = - \left(\frac{\alpha^2}{\eta_1} (de_1 - e_5)^T Z \right. \\ &\quad \times (de_1 - e_5) + \frac{4\alpha^2}{\rho_1} \Phi_3^T (K_2^T K_2 \otimes Z) \Phi_3 \left. \right) \\ \Xi_7 &= -2e_1^T W_3^T H_1 \Gamma_1 H_2 W_3 e_1 + e_1^T W_3^T (H_1 + H_2) \Gamma_1 e_3 \\ &\quad + e_3^T \Gamma_1 (H_1 + H_2) W_3 e_1 - 2e_3^T \Gamma_1 e_3 \\ &\quad - 2e_2^T W_3^T H_1 \Gamma_2 H_2 W_3 e_2 + e_2^T W_3^T (H_1 + H_2) \Gamma_2 e_4 \\ &\quad + e_4^T \Gamma_2 (H_1 + H_2) W_3 e_2 - 2e_4^T \Gamma_2 e_4 - 2[e_3 - e_4 \\ &\quad - H_1 W_3 (e_1 - e_2)]^T \Gamma_3 [e_3 - e_4 - H_2 W_3 (e_1 - e_2)] \\ \rho_0 &= \frac{\eta_0}{\eta_1^2} [\eta_0^2 - (\alpha d)^2 e^{\alpha d}], \quad \eta_0 = e^{\alpha d} - 1, \quad \eta_1 = \eta_0 - \alpha d \\ \rho_1 &= \frac{\eta_1}{\eta_2^2} [2\eta_1^2 - (\alpha d + (\alpha d - 1)\eta_0)(\alpha d)^2] \\ \eta_2 &= \eta_1 - \frac{(\alpha d)^2}{2}, \quad K_1 = \begin{bmatrix} 1 & -\alpha \eta_0 / \eta_1 \end{bmatrix} \\ K_2 &= \begin{bmatrix} 1 & -\alpha \eta_1 / \eta_2 \end{bmatrix} \\ \mathbb{A} &= -C e_1 + W_0 e_3 + W_1 e_4 + W_2 e_6. \end{aligned}$$

Proof: By using the same criteria in Theorem 1, we can obtain the above results. ■

Remark 3: When $W_0 = W_2 = 0$ and $W_1 = I$ in (14), we have the static NN

$$\dot{\varpi}(t) = -C\varpi(t) + f(W_3 \varpi(t-d(t))). \quad (16)$$

When $W_3 = I$ in (14), we have the local-field NN

$$\begin{aligned} \dot{\varpi}(t) &= -C\varpi(t) + W_0 f(\varpi(t)) + W_1 f(\varpi(t-d(t))) \\ &\quad + W_2 \int_{t-\tau(t)}^t f(\varpi(\varepsilon)) d\varepsilon. \end{aligned} \quad (17)$$

Corollary 1: Given $\alpha > 0$ and some scalars $d > 0$, $\mu > 0$, the static NN (16) is exponentially stable when there exist matrices, $P \in \mathcal{R}_{>0}^{3n \times 3n}$ and $Q, R, S, Z \in \mathcal{R}_{>0}^{n \times n}$, and diagonal

matrices, $\Gamma_p \in \mathcal{R}_{>0}^{n \times n}$, ($p = 1, 2, 3$), $\Lambda_q \in \mathcal{R}_{>0}^{n \times n}$ ($q = 1, 2$), Define such that

$$\sum_{m=1}^6 \Xi_m < 0 \quad (18)$$

where

$$\begin{aligned} \Phi_0 &= \begin{bmatrix} e_1 \\ e_5 \\ e_6 \end{bmatrix}, \quad \Phi_1 = \begin{bmatrix} \mathcal{A} \\ e_1 - e_2 \\ de_1 - e_5 \end{bmatrix}, \quad \Phi_2 = \begin{bmatrix} e_1 - e_2 \\ de_1 - e_5 \end{bmatrix} \\ \Phi_3 &= \begin{bmatrix} de_1 - e_5 \\ \frac{d^2}{2}e_1 - e_6 \end{bmatrix}, \quad \Xi_1 = \Phi_0^T P \Theta_1 + \Phi_1^T P \Phi_0 + \alpha \Phi_0^T P \Phi_0 \\ \Xi_2 &= 2[e_3 - H_1 W_3 e_1]^T \Lambda_1 W_3 \mathcal{A} + 2[H_2 W_3 e_1 - e_3]^T \Lambda_2 W_3 \mathcal{A} \\ \Xi_3 &= e_1^T Q e_1 - e^{-\alpha d} e_2^T Q e_2 + e_3^T S e_3 + \pi(\mu) e_4^T S e_4 \\ &\quad + \mathcal{A}^T \left[dR + \frac{d^2}{2} Z \right] \mathcal{A} \\ \Xi_4 &= -\left(\frac{\alpha}{\eta_0} (e_1 - e_2)^T R (e_1 - e_2) + \frac{\alpha}{\rho_0} \Phi_2^T (K_1^T K_1 \otimes R) \Phi_2 \right) \\ \Xi_5 &= -\left(\frac{\alpha^2}{\eta_1} (de_1 - e_5)^T Z \right. \\ &\quad \left. \times (de_1 - e_5) + \frac{4\alpha^2}{\rho_1} \Phi_3^T (K_2^T K_2 \otimes Z) \Phi_3 \right) \\ \Xi_6 &= -2e_1^T W_3^T H_1 \Gamma_1 H_2 W_3 e_1 + e_1^T W_3^T (H_1 + H_2) \Gamma_1 e_3 \\ &\quad + e_3^T \Gamma_1 (H_1 + H_2) W_3 e_1 - 2e_3^T \Gamma_1 e_3 \\ &\quad - 2e_2^T W_3^T H_1 \Gamma_2 H_2 W_3 e_2 + e_2^T W_3^T (H_1 + H_2) \Gamma_2 e_4 \\ &\quad + e_4^T \Gamma_2 (H_1 + H_2) W_3 e_2 - 2e_4^T \Gamma_2 e_4 \\ &\quad - 2[e_3 - e_4 - H_1 W_3 (e_1 - e_2)]^T \Gamma_3 \\ &\quad \times [e_3 - e_4 - H_2 W_3 (e_1 - e_2)], \quad K_1 = \begin{bmatrix} 1 & -\alpha\eta_0/\eta_1 \end{bmatrix} \\ K_2 &= \begin{bmatrix} 1 & -\alpha\eta_1/\eta_2 \end{bmatrix}, \quad \mathcal{A} = -Ce_1, \quad \eta_0 = e^{\alpha d} - 1 \\ \eta_1 &= \eta_0 - \alpha d, \quad \eta_2 = \eta_1 - \frac{(\alpha d)^2}{2}, \quad \rho_0 = \frac{\eta_0}{\eta_1^2} \left[\eta_0^2 - (\alpha d)^2 e^{\alpha d} \right] \\ \rho_1 &= \frac{\eta_1}{\eta_2^2} \left[2\eta_1^2 - (\alpha d + (\alpha d - 1)\eta_0)(\alpha d)^2 \right]. \end{aligned}$$

Proof: Choose the following LKF:

$$\begin{aligned} V_1(\varpi_t) &= \vartheta^T(t) P \vartheta(t) + 2 \sum_{i=1}^n \lambda_{1i} \int_0^{W_{3i}\varpi(t)} (f_i(\varepsilon) - m_i^- \varepsilon) d\varepsilon \\ &\quad + 2 \sum_{i=1}^n \lambda_{2i} \int_0^{W_{3i}\varpi(t)} (m_i^+ \varepsilon - f_i(\varepsilon)) d\varepsilon \\ &\quad + \int_{t-d}^t e^{\alpha(s-t)} \varpi^T(\varepsilon) Q \varpi(\varepsilon) d\varepsilon \\ &\quad + \int_{-d}^0 \int_{t+\varepsilon}^t e^{\alpha(\theta-t)} \dot{\varpi}^T(\theta) R \dot{\varpi}(\theta) d\theta d\varepsilon \\ &\quad + \int_{t-d(t)}^t e^{\alpha(s-t)} f^T(W_3 \varpi(\varepsilon)) S f(W_3 \varpi(\varepsilon)) d\varepsilon \\ &\quad + \int_{-d}^0 \int_{\varepsilon}^0 \int_{t+\theta}^t e^{\alpha(u-t)} \dot{\varpi}^T(u) Z \dot{\varpi}(u) du d\theta d\varepsilon. \quad (19) \end{aligned}$$

$$\begin{aligned} e_j &= \begin{bmatrix} 0_{n \times (j-1)n} & I_n & 0_{n \times (6-j)n} \end{bmatrix}, \quad j = 1, 2, \dots, 6 \\ \xi_1(t) &= \begin{bmatrix} \varpi^T(t) & \varpi^T(t-d) & f(W_3 \varpi(t)) \\ \times f^T(W_3 \varpi(t-d(t))) & \int_{t-d}^t \varpi^T(\varepsilon) d\varepsilon \\ \times \int_{-d}^0 \int_{t+\varepsilon}^t \varpi^T(u) du d\varepsilon \end{bmatrix}^T. \end{aligned}$$

By using an approach similar to that used in Theorem 1, we can obtain the exponential stability conditions in Corollary 1. ■

When $W_2 = 0$, NN (17) becomes

$$\dot{\varpi}(t) = -C\varpi(t) + W_0 f(\varpi(t)) + W_1 f(\varpi(t-d(t))). \quad (20)$$

In what follows, a sufficient condition is obtained for the NN (20).

Corollary 2: Given $\alpha > 0$ and some scalars $d > 0$, $\mu > 0$, the GNN (20) is exponentially stable when there exist matrices $P \in \mathcal{R}_{>0}^{3n \times 3n}$ and $Q, R, S, T, Z \in \mathcal{R}_{>0}^{n \times n}$, and diagonal matrices, $\Gamma_p \in \mathcal{R}_{>0}^{n \times n}$, ($p = 1, 2, 3$), $\Lambda_q \in \mathcal{R}_{>0}^{n \times n}$ ($q = 1, 2$), such that

$$\sum_{m=1}^6 \Xi_m < 0 \quad (21)$$

where

$$\begin{aligned} \Phi_0 &= \begin{bmatrix} e_1 \\ e_5 \\ e_6 \end{bmatrix}, \quad \Phi_1 = \begin{bmatrix} \bar{\mathcal{A}} \\ e_1 - e_2 \\ de_1 - e_5 \end{bmatrix}, \quad \Phi_2 = \begin{bmatrix} e_1 - e_2 \\ de_1 - e_5 \end{bmatrix} \\ \Phi_3 &= \begin{bmatrix} de_1 - e_5 \\ \frac{d^2}{2}e_1 - e_6 \end{bmatrix}, \quad \Xi_1 = \Phi_0^T P \Theta_1 + \Phi_1^T P \Phi_0 + \alpha \Phi_0^T P \Phi_0 \\ \Xi_2 &= 2[e_3 - H_1 e_1]^T \Lambda_1 \bar{\mathcal{A}} + 2[H_2 e_1 - e_3]^T \Lambda_2 \bar{\mathcal{A}} \\ \Xi_3 &= e_1^T Q e_1 - e^{-\alpha d} e_2^T Q e_2 + e_3^T S e_3 + \pi(\mu) e_4^T S e_4 \\ &\quad + \bar{\mathcal{A}}^T \left[dR + \frac{d^2}{2} Z \right] \bar{\mathcal{A}} \\ \Xi_4 &= -\left(\frac{\alpha}{\eta_0} (e_1 - e_2)^T R (e_1 - e_2) + \frac{\alpha}{\rho_0} \Phi_2^T (K_1^T K_1 \otimes R) \Phi_2 \right) \\ \Xi_5 &= -\left(\frac{\alpha^2}{\eta_1} (de_1 - e_5)^T Z \right. \\ &\quad \left. \times (de_1 - e_5) + \frac{4\alpha^2}{\rho_1} \Phi_3^T (K_2^T K_2 \otimes Z) \Phi_3 \right) \\ \Xi_6 &= -2e_1^T H_1 \Gamma_1 H_2 e_1 + e_1^T \Gamma_1 (H_1 + H_2) e_3 \\ &\quad + e_3^T (H_1 + H_2) \Gamma_2 e_1 - 2e_3^T \Gamma_1 e_3 \\ &\quad - 2e_2^T H_1 \Gamma_2 H_2 e_2 + e_2^T \Gamma_2 (H_1 + H_2) e_4 \\ &\quad + e_4^T (H_1 + H_2) \Gamma_2 e_2 - 2e_4^T \Gamma_2 e_4 \\ \rho_0 &= \frac{\eta_0}{\eta_1^2} \left[\eta_0^2 - (\alpha d)^2 e^{\alpha d} \right] \quad \eta_0 = e^{\alpha d} - 1, \quad \eta_1 = \eta_0 - \alpha d \\ \rho_1 &= \frac{\eta_1}{\eta_2^2} \left[2\eta_1^2 - (\alpha d + (\alpha d - 1)\eta_0)(\alpha d)^2 \right] \\ \eta_2 &= \eta_1 - \frac{(\alpha d)^2}{2}, \quad K_1 = \begin{bmatrix} 1 & -\alpha\eta_0/\eta_1 \end{bmatrix} \\ K_2 &= \begin{bmatrix} 1 & -\alpha\eta_1/\eta_2 \end{bmatrix}, \quad \bar{\mathcal{A}} = -Ce_1 + W_0 e_3 + W_2 e_4. \end{aligned}$$

Proof: Choose the LKF as follows:

$$\begin{aligned}
V_2(\varpi_t) = & \vartheta^T(t)P\vartheta(t) + 2 \sum_{i=1}^n \lambda_{1i} \int_0^{W_{3i}\varpi(t)} (f_i(\varepsilon) - m_i^-\varepsilon) d\varepsilon \\
& + 2 \sum_{i=1}^n \lambda_{2i} \int_0^{W_{3i}\varpi(t)} (m_i^+\varepsilon - f_i(\varepsilon)) d\varepsilon \\
& + \int_{t-d}^t e^{\alpha(s-t)} \varpi^T(\varepsilon) Q \varpi(\varepsilon) d\varepsilon \\
& + \int_{-d}^0 \int_{t+\varepsilon}^t e^{\alpha(\theta-t)} \dot{\varpi}^T(\theta) R \dot{\varpi}(\theta) d\theta d\varepsilon \\
& + \int_{t-d(t)}^t e^{\alpha(s-t)} f^T(W_3\varpi(\varepsilon)) S f(W_3\varpi(\varepsilon)) d\varepsilon \\
& + \int_{-d}^0 \int_{\varepsilon}^t \int_{t+\theta}^t e^{\alpha(u-t)} \dot{\varpi}^T(u) Z \dot{\varpi}(u) du d\theta d\varepsilon. \quad (22)
\end{aligned}$$

Let $\Gamma_p = \text{diag}\{\eta_{p1}, \eta_{p2}, \dots, \eta_{pn}\} > 0$ and $(p = 1, 2)$ be positive-diagonal matrices. Then, from Assumption 1, we have

$$0 \leq -2f^T(\varpi(t))\Gamma_1 f(\varpi(t)) + 2\varpi^T(t)\Gamma_1(H_1 + H_2)f(\varpi(t)) - 2\varpi^T(t)H_1\Gamma_1H_2\varpi(t) \quad (23)$$

$$\begin{aligned}
0 \leq & -2f^T(\varpi(t - \tau(t)))\Gamma_2 f(\varpi(t - \tau(t))) + 2\varpi^T(t - \tau(t)) \\
& \times \Gamma_2(H_1 + H_2)f(\varpi(t - \tau(t))) \\
& - 2\varpi^T(t - \tau(t))H_1\Gamma_2H_2\varpi(t - \tau(t)). \quad (24)
\end{aligned}$$

Define

$$\begin{aligned}
e_j = & [0_{n \times (j-1)n} \ I_n \ 0_{n \times (6-j)n}], \ j = 1, 2, \dots, 6 \\
\xi_2(t) = & \begin{bmatrix} \varpi^T(t) & \varpi^T(t-d) & f(\varpi(t)) & f^T(\varpi(t-d(t))) \\ \times \int_{t-d}^t \varpi^T(\varepsilon) d\varepsilon & \int_{-d}^0 \int_{t+\varepsilon}^t \varpi^T(u) du d\varepsilon \end{bmatrix}^T.
\end{aligned}$$

By using an approach similar to that used in Theorem 1, we can obtain the above exponential stability conditions. ■

Remark 4: Recently, GNNs have attracted great research interest; for example, the stability analysis problem of GNNs including additive time-varying delayed signals was examined in [30]. In [31], the improved stability of GNNs including interval time-varying delayed states was studied. The H_∞ state estimation of GNNs including interval time-delayed signals was investigated in [32]. Very recently, in [33], the stability problem was revisited for Markovian jump delayed GNNs. In the above methods, the RCC and Wirtinger-based double integral inequality (WDI) were utilized to obtain less conservative results. Thus, the above two techniques are the best methods for reducing conservatism. However, the exponential stability analysis problem for delayed GNNs has not been completely considered in the above results. In addition, the RCC and WDI techniques may not be accurate for exponential stability problems. In this paper, we utilized the new WII for both single and double integral terms.

IV. NUMERICAL EXAMPLES

Example 1: Let us consider the following matrix values:

$$C = \begin{bmatrix} 6.5618 & 0 & 0 \\ 0 & 5.5784 & 0 \\ 0 & 0 & 7.3269 \end{bmatrix}$$

$$\begin{aligned}
W_0 = & \begin{bmatrix} 0.3256 & -0.1904 & 0.3322 \\ -0.1564 & 0.2446 & 0.3674 \\ -0.1753 & 0.2956 & -0.3115 \end{bmatrix} \\
W_1 = & \begin{bmatrix} 0.1981 & -0.1313 & 0.1185 \\ 0.1645 & 0.0901 & 0.1013 \\ 0.0274 & -0.1518 & 0.0742 \end{bmatrix} \\
W_2 = & \begin{bmatrix} 1.8647 & 0.7983 & -0.8192 \\ 0.6983 & 1.4083 & 1.2748 \\ -0.4132 & -0.6093 & 1.2183 \end{bmatrix} \\
W_3 = & \begin{bmatrix} 0.4631 & 1.7363 & -1.7335 \\ 1.8369 & 1.4830 & 1.7428 \\ -1.4521 & -0.9603 & 1.8736 \end{bmatrix} \\
H_1 = & \text{diag}\{0, 0, 0\}, \ H_2 = \text{diag}\{0.5, 0.5, 0.5\}.
\end{aligned}$$

A. Stability Analysis

By considering the GNN (14) with $\mu = 1.2$, $d = 1$, and $\tau = 0.5$ and applying Theorem 2, we can obtain the following feasible solutions to LMI (15):

$$\begin{aligned}
P = & \begin{bmatrix} 39.0691 & -18.5234 & 20.7314 & 1.9127 & -1.1049 \\ -18.5234 & 45.6462 & -3.5469 & -1.3266 & 1.2421 \\ 20.7314 & -3.5469 & 63.9502 & 2.8042 & -0.5195 \\ 1.9127 & -1.3266 & 2.8042 & 13.6448 & -3.7799 \\ -1.1049 & 1.2421 & -0.5195 & -3.7799 & 15.2429 \\ 2.6739 & -1.6862 & 5.3377 & 8.2933 & -3.6949 \\ 7.9196 & -4.9988 & 5.7114 & -2.0523 & -2.1909 \\ -5.5125 & 7.1939 & -1.7188 & -1.8112 & -5.0803 \\ 7.3552 & -3.5224 & 18.1741 & 1.1287 & -0.6741 \\ 2.6739 & 7.9196 & -5.5125 & 7.3552 & \\ -1.6862 & -4.9988 & 7.1939 & -3.5224 & \\ 5.3377 & 5.7114 & -1.7188 & 18.1741 & \\ 8.2933 & -2.0523 & -1.8112 & 1.1287 & \\ -3.6949 & -2.1909 & -5.0803 & -0.6741 & \\ 26.4836 & 0.5188 & -0.6916 & 1.9064 & \\ 0.5188 & 46.4903 & -13.2618 & 16.8349 & \\ -0.6916 & -13.2618 & 43.8566 & -6.1617 & \\ 1.9064 & 16.8349 & -6.1617 & 79.2727 & \end{bmatrix} \\
Q = & \begin{bmatrix} 114.3368 & -6.6495 & 2.7397 \\ -6.6495 & 90.7478 & -9.4362 \\ 2.7397 & -9.4362 & 161.1047 \end{bmatrix} \\
R = & \begin{bmatrix} 7.7549 & -1.1358 & 4.4495 \\ -1.1358 & 10.4849 & -1.8651 \\ 4.4495 & -1.8651 & 13.6709 \end{bmatrix} \\
S = & \begin{bmatrix} 20.7488 & -11.5626 & 5.8076 \\ -11.5626 & 19.0814 & 0.8206 \\ 5.8076 & 0.8206 & 26.6407 \end{bmatrix} \\
T = & \begin{bmatrix} 348.7732 & 136.4528 & 27.2418 \\ 136.4528 & 332.9313 & 66.3766 \\ 27.2418 & 66.3766 & 359.9980 \end{bmatrix} \\
Z = & \begin{bmatrix} 3.0029 & -0.4653 & 0.5172 \\ -0.4653 & 3.0143 & -0.1786 \\ 0.5172 & -0.1786 & 4.0146 \end{bmatrix}, \quad \alpha = 1.7768.
\end{aligned}$$

Therefore, the GNN (14) attains exponential stability. When $\tau = 0.3$, the maximum exponential convergence rates $\alpha_{\max}$ are acquired for various d and μ , which are given in Table I. Table II shows the $\alpha_{\max}$ for various d and τ when $d(t)$ is not differentiable.

TABLE I
MAXIMUM EXPONENTIAL CONVERGENCE RATES
 $\alpha_{\max}$ FOR DIFFERENT d AND μ

$d \backslash \mu$	0.1	0.3	0.6	0.9	1.2
0.5	5.3278	5.0984	4.9107	4.7387	4.5536
1.0	3.2214	3.1845	2.9629	2.6867	2.3849
1.5	2.1358	2.0735	1.9334	1.7635	1.5178
2.0	1.6874	1.5684	1.4732	1.3861	1.2814

TABLE II
MAXIMUM EXPONENTIAL CONVERGENCE RATES
 $\alpha_{\max}$ FOR DIFFERENT d AND τ

$d \backslash \tau$	0.1	0.3	0.5	0.7	0.9
0.5	6.7825	6.4684	4.7756	3.4654	2.4354
1.0	4.4342	3.1354	2.3565	1.7347	1.2267
1.5	3.1567	2.1731	1.5513	1.1613	0.8167
2.0	2.3534	1.6518	1.2021	0.8535	0.6196

TABLE III
MINIMUM PASSIVITY INDEX γ FOR DIFFERENT VALUES OF d

$d \backslash \gamma$	0.5	1	1.5	2	2.5
Theorem 3.1	0.3035	0.3463	0.3648	0.3847	0.4258

TABLE IV
OPTIMAL DISSIPATIVITY INDEX γ FOR DIFFERENT VALUES OF d

$d \backslash \gamma$	0.5	1	1.5	2	2.5
Theorem 3.1	1.7324	1.7215	1.7017	1.6814	1.6583

B. Passivity Analysis

When $\mathfrak{Q} = 0$, $\mathfrak{S} = I_3$, and $\mathfrak{R} = \gamma I_3$, the passivity conditions are analyzed for the GNN (1) by applying Theorem 1. By taking $\mu = 1.2$ and $\tau = 0.5$, the minimum passivity index γ is obtained for different values of d (see Table III).

C. $(\mathfrak{Q}, \mathfrak{S}, \mathfrak{R})$ - γ -Dissipativity Analysis

When $\mathfrak{Q} = 0.5I_3$, $\mathfrak{S} = I_3$, and $\mathfrak{R} = 2I_3 - \gamma I_3$, the dissipativity conditions are analyzed for the GNN (1) by applying Theorem 1. By taking $\mu = 1.2$ and $\tau = 0.5$, the optimal dissipativity index γ is obtained for different values of d (see Table IV).

Example 2: Let us consider a GNN (17) with the parameters given in [39]

$$C = \text{diag}\{2.3, 3.4, 2.5\}, \quad W_0 = \begin{bmatrix} 0.9 & -1.5 & 0.1 \\ -1.2 & 1 & 0.2 \\ 0.2 & 0.3 & 0.8 \end{bmatrix}$$

$$W_1 = \begin{bmatrix} 0.8 & 0.6 & 0.2 \\ 0.5 & 0.7 & 0.1 \\ 0.2 & 0.1 & 0.5 \end{bmatrix}, \quad W_2 = \begin{bmatrix} 0.3 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.1 \\ 0.1 & 0.1 & 0.2 \end{bmatrix}$$

$$H_1 = 0, \quad H_2 = 0.2I_3.$$

When $d = 1$, by applying Theorem 2, the maximum allowable upper bound (MAUB) τ is obtained for different values of α (see Table V). The number of decision variables (NDV) in [39] is $((25n(n+1))/2) + 6n^2$, and the NDV in Theorem 2 is $((13n(n+1))/2) + 3n^2$. Therefore, the approach of this paper provides less conservative results than [39]. The state trajectories of the state $\varpi(t)$ is shown in Fig. 1.

TABLE V
MAUB τ FOR DIFFERENT VALUES OF α

$\tau \backslash \alpha$	0.1	0.3	0.5	0.8
Ref. [40]	9.127	4.895	3.421	2.313
Theorem 3.3	11.654	6.967	5.762	4.683

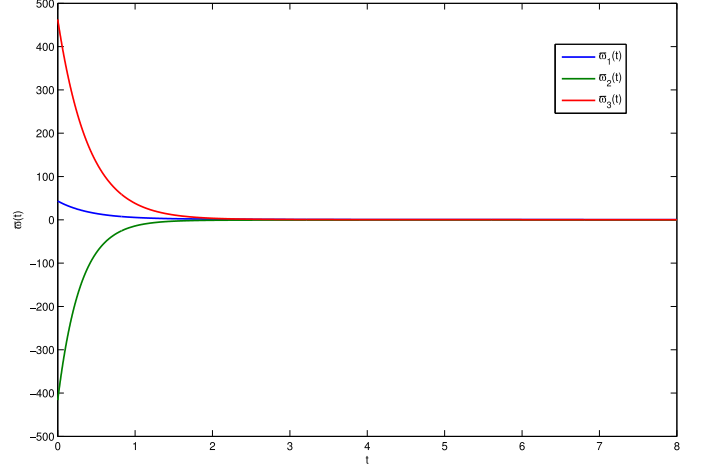


Fig. 1. State response $\varpi(t)$ in Example 2.

TABLE VI
MAXIMUM ADMISSIBLE CONVERGENCE RATES
 $\alpha_{\max}$ FOR DIFFERENT d , τ , AND μ

Methods	[37]	[38]	[39]	[40]	This paper
$d = 0.8, \tau = 0.2, \mu = 0.5$	0.10	0.11	0.30	0.21	0.43
$d = 0.6, \tau = 0.2, \mu = 0.5$	0.33	0.29	0.55	0.42	1.00
$d = 0.5, \tau = 0.2, \mu = 0.5$	0.35	0.45	0.73	0.59	1.39
$d = 0.5, \tau = 0.2, \mu = 0$	0.56	0.67	0.86	0.89	1.76

Example 3: Consider an NN (17) with the following parameters:

$$C = \text{diag}\{6, 5, 7\}, \quad W_0 = \begin{bmatrix} 1.2 & -0.8 & 0.6 \\ 0.5 & -1.5 & 0.7 \\ -0.8 & -1.2 & -1.4 \end{bmatrix}$$

$$W_1 = \begin{bmatrix} -1.4 & 0.9 & 0.5 \\ -0.6 & 1.2 & 0.8 \\ 0.5 & -0.7 & 1.1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 1.8 & 0.7 & -0.8 \\ 0.6 & 1.4 & 1.0 \\ -0.4 & -0.6 & 1.2 \end{bmatrix}.$$

By applying Theorem 2, we obtain the maximum admissible convergence rate $\alpha_{\max}$. Table VI lists the maximum admissible convergence rates $\alpha_{\max}$ for different values of d , τ , and μ when $H_1 = \text{diag}\{-1.2, 0, -2.4\}$ and $H_2 = \{0, 1.4, 0\}$. Table VII lists the maximum admissible convergence rates $\alpha_{\max}$ for different values of μ when $d = \tau = 0.2$, $H_1 = \text{diag}\{0, 0, 0\}$, and $H_2 = \{-1.2, 1.4, -2.4\}$. Table VIII lists the maximum admissible convergence rates $\alpha_{\max}$ for different values of τ when $d = \mu = 0.2$, $H_1 = \text{diag}\{0, 0, 0\}$, and $H_2 = \{-1.2, 1.4, -2.4\}$. The NDVs in [36]–[41] are $((17n(n+1))/2) + 2n^2$, $((16n(n+1))/2) + 32n^2$, $((19n(n+1))/2) + 7n^2$, $((25n(n+1))/2) + 6n^2$, $((9n(n+1))/2)$, and

TABLE VII
MAXIMUM ADMISSIBLE CONVERGENCE RATES $\alpha_{\max}$
FOR DIFFERENT VALUES OF μ AND $\tau = 0.2$

$d = 0.2$	[41]	[42]	This paper
$\mu = 0.1$	0.76	1.48	3.24
$\mu = 0.3$	0.70	1.34	2.87
$\mu = 0.5$	0.67	1.18	2.54
unknown $\mu = 0.5$	0.64	0.78	1.96

TABLE VIII
MAXIMUM ADMISSIBLE CONVERGENCE RATES $\alpha_{\max}$
FOR DIFFERENT VALUES OF τ AND $d = 0.2, \mu = 0.2$

$d = 0.2$	[41]	[42]	This paper
$\tau = 0.1$	1.00	1.54	3.35
$\tau = 0.2$	0.73	1.41	2.98
$\tau = 0.3$	0.49	1.23	2.65

TABLE IX
MAXIMUM ADMISSIBLE CONVERGENCE RATES $\alpha_{\max}$
FOR VARIOUS VALUES OF μ WITH $d = 1$

$\alpha \backslash \mu$	0.0	0.1	0.2	unknown μ
Ref. [30]	0.2505	0.2378	0.2273	-
Corollary 3.5	2.3546	2.1954	2.0769	1.8769

$((13n(n+1))/2) + 6n^2$, respectively. The differences in the NDVs between the methods in [36]–[41] and Theorem 2 are $((4n(n+1))/2) - n^2$, $((3n(n+1))/2) + 29n^2$, $((6n(n+1))/2) + 4n^2$, $((12n(n+1))/2) + 3n^2$, $-((4n(n+1))/2) - 3n^2$, and $3n^2$, respectively. This demonstrates that the method established in this paper is less conservative than the methods given in [36]–[41].

Example 4: Choose the NN (16) with the following matrix values in [29]:

$$C = \text{diag}\{4.1989, 0.7160, 1.9985\}$$

$$W_3 = \begin{bmatrix} -0.1052 & -0.5069 & -0.1121 \\ -0.0257 & -0.2808 & 0.0212 \\ 0.1205 & -0.2153 & 0.1315 \end{bmatrix}$$

$$H_1 = \text{diag}\{0, 0, 0\}, H_2 = \{0.4219, 3.8993, 1.0160\}.$$

Table IX lists the $\alpha_{\max}$ for various values of μ when $d = 1$. The NDV in [29] is $((12n(n+1))/2) + 10n^2$, while that in Corollary 1 is $((12n(n+1))/2) + 3n^2$.

Example 5: Choose the NN (17) with the following parameters:

$$C = \begin{bmatrix} 3.99 & 0 \\ 0 & 2.99 \end{bmatrix}, W_1 = \begin{bmatrix} 0.009 & 0.14 \\ 0.05 & 0.09 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 0.45 & -0.2 \\ 0.3 & 0.42 \end{bmatrix}, W_3 = I_2$$

$$H_1 = \text{diag}\{0, 0\}, H_2 = \{1, 1\}.$$

Table X lists the MAUB d for various values of μ with $\alpha = 0.5, \tau = 0.1$ by using Theorem 2. It demonstrates that the method presented in this paper provides results with reduced conservatism compared with [36] and [39]. The state trajectories of the state $\varpi(t)$ is shown in Fig. 2.

TABLE X
MAUB d FOR VARIOUS μ WITH $\alpha = 0.5, \tau = 0.1$

$d \backslash \mu$	0.0	0.2	0.5	0.7
Ref. [37]	3.377	3.336	3.268	3.264
Ref. [40]	3.415	3.395	3.356	3.325
Theorem 3.3	5.346	4.876	4.592	4.264

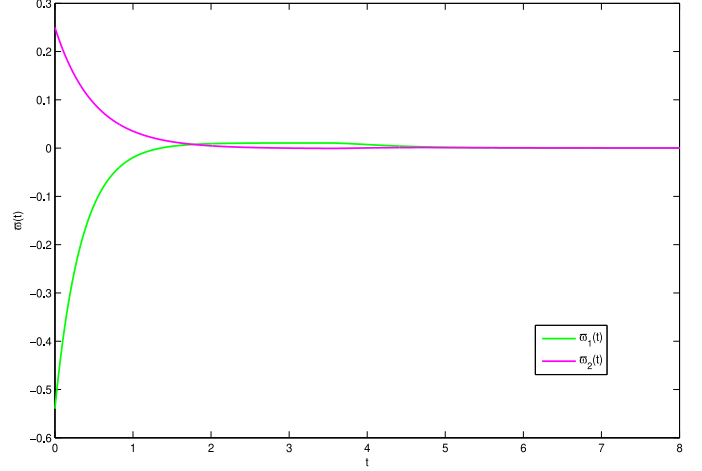


Fig. 2. State response $\varpi(t)$ in Example 5.

TABLE XI
 $\alpha_{\max}$ FOR VARIOUS μ

$\alpha \backslash \mu$	0.0	0.2	0.5	0.7
Ref. [62]	1.1540	0.8696	0.8354	0.8169
Ref. [59]	1.1544	0.8784	0.8484	0.8217
Corollary 3.6	1.6849	1.2893	1.2109	1.1938

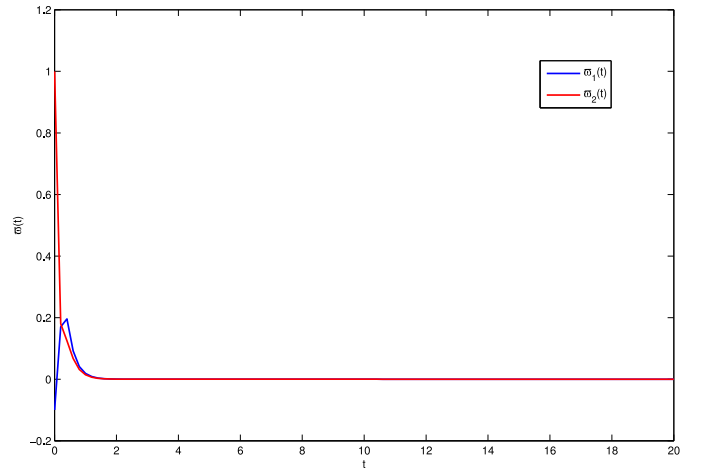


Fig. 3. State response $\varpi(t)$ in Example 6.

Example 6: Choose the NN (20) with the following parameters:

$$C = \begin{bmatrix} 2 & 0 \\ 0 & 3.5 \end{bmatrix}, W_0 = \begin{bmatrix} -1 & 0.5 \\ 0.5 & -1 \end{bmatrix}, W_1 = \begin{bmatrix} -0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

$$H_1 = \text{diag}\{0, 0\}, H_2 = \{1, 1\}.$$

By applying Corollary 2, we obtain the maximum allowable exponential convergence rates $\alpha_{\max}$ for various values of μ

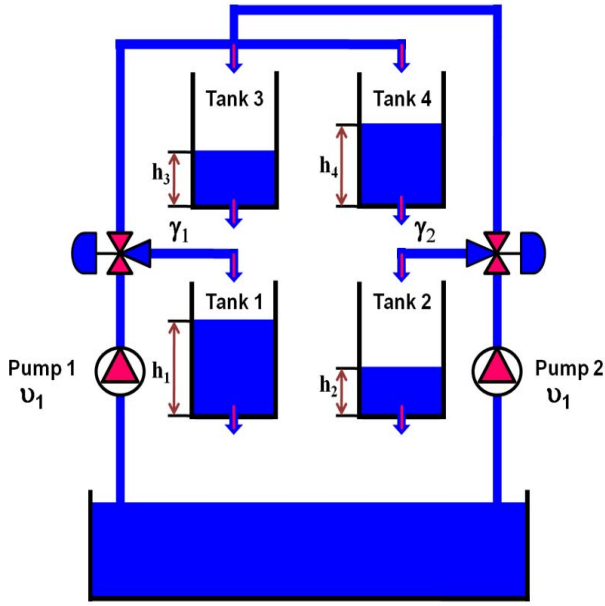


Fig. 4. Schematic representation of QTP (source [62]).

when $d = 1$, which are listed in Table XI. The state trajectories of the state $\varpi(t)$ is shown in Fig. 3.

Example 7: To show the application of artificial NNs, the quadruple-tank process (QTP) (see Fig. 4) has been chosen in this example. By using pumps 1 and 2, the target is to stabilize the liquid level in tanks 1 and 2, respectively. In the QTP, v_1 and v_2 are the input voltages to pumps 1 and 2, respectively. The outputs of pumps 1 and 2 are γ_1 and γ_2 (voltages from level measurement devices), respectively. The QTP has been expressed in the following system (see [62]):

$$\dot{\hat{x}}(t) = \hat{A}_0 \hat{x}(t) + \hat{A}_1 \hat{x}(t - d_1) + \hat{B}_0 \hat{u}(t - d_2) + \hat{B}_1 \hat{u}(t - d_3) \quad (25)$$

where

$$\begin{aligned} \hat{A}_0 &= \text{diag}\{-0.0021, -0.0021, -0.0424, -0.0424\} \\ \hat{A}_1 &= \begin{bmatrix} 0 & 0 & 0.0424 & 0 \\ 0 & 0 & 0 & 0.0424 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ \hat{B}_0 &= \begin{bmatrix} 0.1113\gamma_1 & 0 & 0 & 0 \\ 0 & 0.1042\gamma_2 & 0 & 0 \end{bmatrix} \\ \hat{B}_1 &= \begin{bmatrix} 0 & 0 & 0 & 0.1113\gamma_1 \\ 0 & 0 & 0.1042\gamma_2 & 0 \end{bmatrix} \\ \hat{K} &= \begin{bmatrix} -0.1609 & -0.1765 & -0.0795 & -0.2073 \\ -0.1977 & -0.1579 & -0.2288 & -0.0772 \end{bmatrix} \\ \gamma_1 &= 0.333, \quad \gamma_2 = 0.307, \quad \hat{u}(t) = \hat{K}\hat{x}(t). \end{aligned}$$

Usually, by using differential equations the mass balances can be expressed. The transport delays are generally included with the delay phenomenon in the inlet of water to the tanks to obtain an interesting control problem. In addition, in this example transport delays between the valves and tanks are chosen as time-varying, and it is assumed that $d_1 = 0$, $d_2 = 0$, and $d_3 = d(t)$. The control input variable $\hat{u}(t)$ denotes the amount of water provided by the pumps. Therefore, it is easy

TABLE XII
UPPER BOUNDS OF d FOR VARIOUS VALUES OF γ_1 AND γ_2

$\gamma_2 \backslash \gamma_1$	0.3	0.4	0.5
0.3	0.9845	1.0058	1.0954
0.4	1.0084	1.0656	1.1072
0.5	1.0987	1.1193	1.1209

to see that $\hat{u}(t)$ exhibits a threshold value because of the limited capacity area and pump hoses. Thus, in this example the nonlinear function $\hat{u}(t)$ is considered as follows:

$$\begin{aligned} \hat{u}(t) &= \hat{K}\hat{f}(\hat{x}(t)) \\ \hat{f}(\hat{x}(t)) &= [\hat{f}_1(\hat{x}_1(t)), \dots, \hat{f}_4(\hat{x}_4(t))]^T \\ \hat{f}_i(\hat{x}_i(t)) &= 0.01(|\hat{x}_i(t) + 1| - |\hat{x}_i(t) - 1|), \quad i = 1, \dots, 4. \end{aligned}$$

The QTP (25) can be transferred to the GNN (1), with $W_3 = I$ and $W_2 = \tau = 0$, as follows:

$$\begin{aligned} \dot{\varpi}(t) &= -C\varpi(t) + W_0 f(\varpi(t)) + W_1 f(\varpi(t - d(t))) + J \\ \mathfrak{z}(t) &= f(\varpi(t)) \end{aligned} \quad (26)$$

where $C = -\hat{A}_0 - \hat{A}_1$, $W_0 = \hat{B}_0 \hat{K}$, $W_1 = \hat{B}_1 \hat{K}$, $f(\cdot) = \hat{f}(\cdot)$. Additionally, by taking the values

$$J = [0 \quad 0 \quad 0 \quad 0]^T, \quad H_1 = 0_4 \quad \text{and} \quad H_2 = 0.01I_4$$

the delay bound $d = 0.9867$ has been obtained; therefore, the LMI in Theorem 1 is feasible. In addition, the upper bounds of d for various values of γ_1 and γ_2 are given in Table XII.

Remark 5: The works in [38], [39], and [41] were restricted to the assumption that the derivative of the time-varying delay is either bounded or less than 1. However, this assumption is not required in this paper.

Remark 6: By establishing a suitable LKF and using the WII technique, the exponential stability, passivity, and $(\Omega, \mathfrak{S}, \mathfrak{R})$ - γ -dissipativity conditions are obtained without using any free-weighting matrices. To avoid numerical computational complexity, the LKF established in this paper has fewer NDVs than the methods in [36]–[39], [41], and [58]. In addition, through the new analysis method, we obtained the MAUBs. Therefore, the obtained results are less conservative than the approaches in [36]–[39], [41], and [58].

V. CONCLUSION

In this paper, we have examined the exponential stability, passivity, and $(\Omega, \mathfrak{S}, \mathfrak{R})$ - γ -dissipativity problems for delayed GNNs including discrete and distributed time-varying delayed states. Improved exponential stability, passivity, and $(\Omega, \mathfrak{S}, \mathfrak{R})$ - γ -dissipativity conditions were established by choosing a suitable LKF. The proposed conditions were represented in the form of LMIs by using a new WII technique for both single and double integrals and a new mathematical analysis method. Six numerical examples were used to explain the superiority of the presented results. The obtained results were compared with those of recent methods in the literature. We wish to extend this paper to the stochastic exponential stability, passivity, and $(\Omega, \mathfrak{S}, \mathfrak{R})$ - γ -dissipativity problems for GNNs including Markovian jumping parameters.

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